dynamically at runtime, allowing reuse of procedures and functions for specific problems, such as OpenGL for 3D graphics, or libraries for TCP/IP networking. Such APIs are still common, but a growing number of APIs are now made available over the internet as RESTful web services.

Broadly speaking, an API is a boundary between one part of a software system and another. It defines a set of operations that one part of the system provides for other parts of the system (or other systems) to make use of. For example, a photography archive might provide an API to list albums of photos, to view individual photos, add comments, and so on. An online image gallery could then use that API to display interesting photos, while a word processor application could use the same API to allow embedding images into a document. As shown in figure 1.1, an API handles requests from one or more clients on behalf of users. A client may be a web or mobile application with a user interface (UI), or it may be another API with no explicit UI. The API itself may talk to other APIs to get its work done.

